

E9 205 Machine Learning for Signal Processing

Convolutional and Recurrent Networks

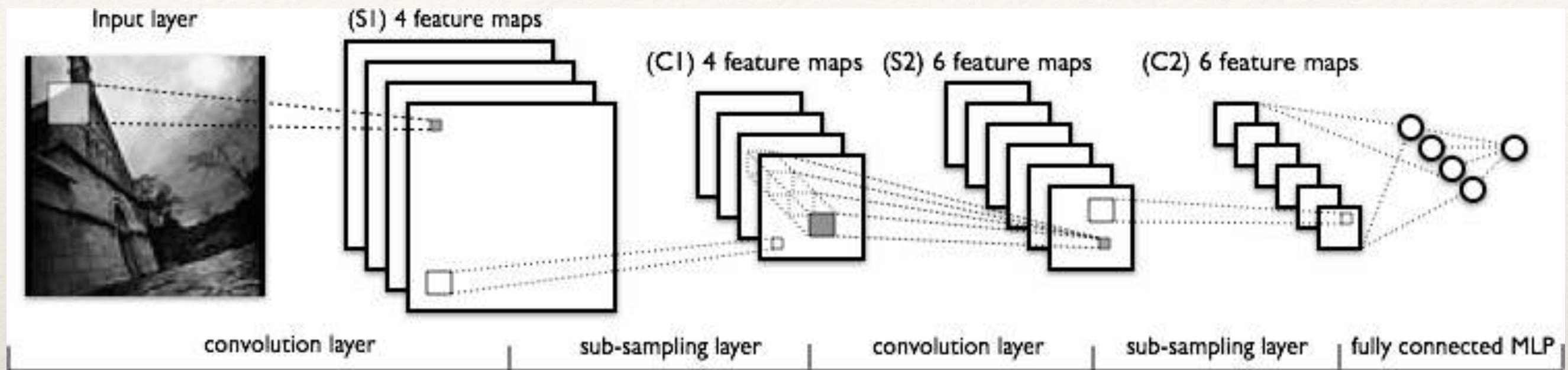
18-11-2017

Bias Variance Trade Off and Overfitting

$$(\text{bias})^2 = \frac{1}{2} \int \{\mathcal{E}_D[y(\mathbf{x})] - \langle t|\mathbf{x} \rangle\}^2 p(\mathbf{x}) d\mathbf{x}$$

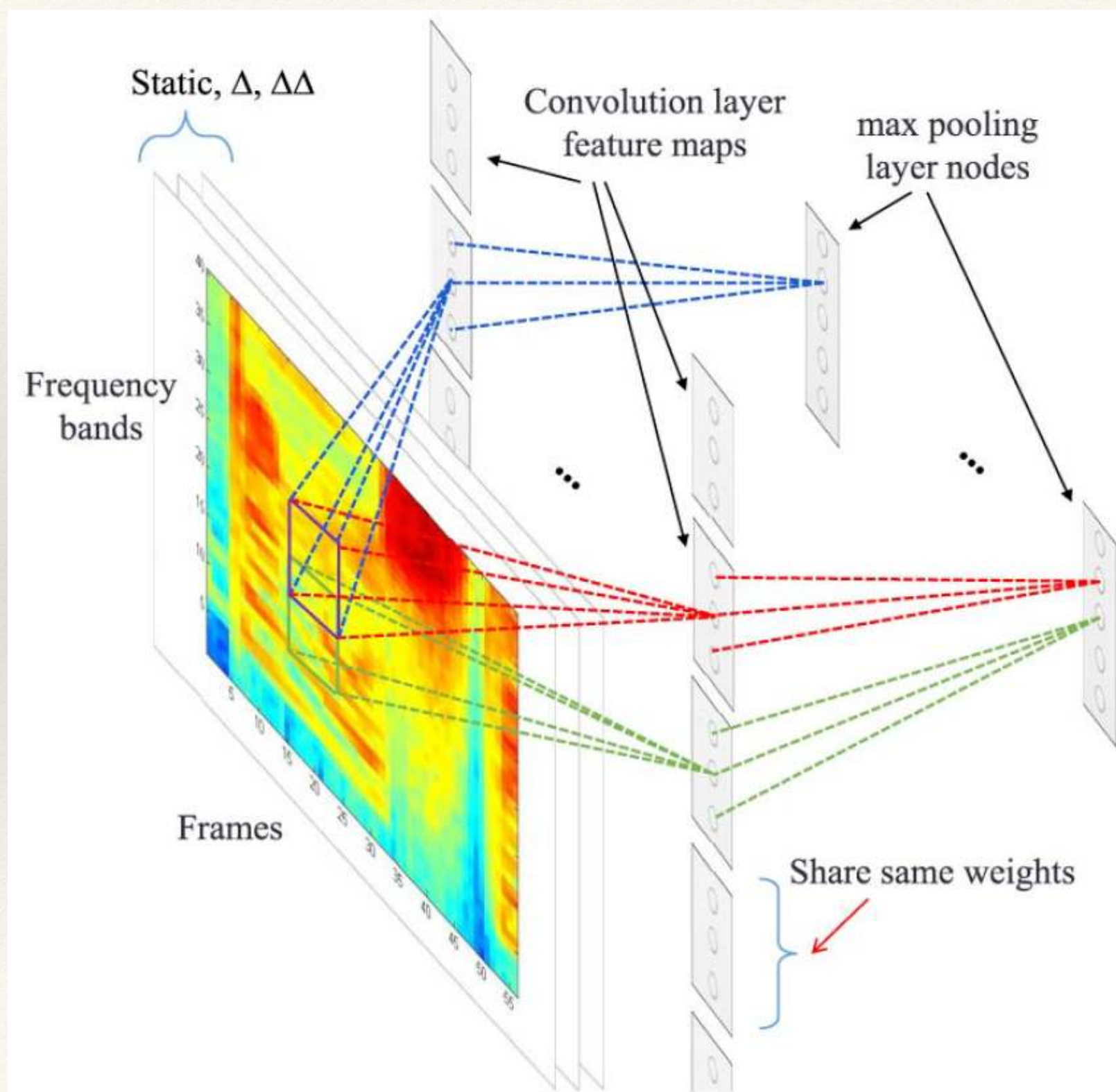
$$\text{variance} = \frac{1}{2} \int \mathcal{E}_D[\{y(\mathbf{x}) - \mathcal{E}_D[y(\mathbf{x})]\}^2] p(\mathbf{x}) d\mathbf{x}.$$

Convolutional Neural Networks



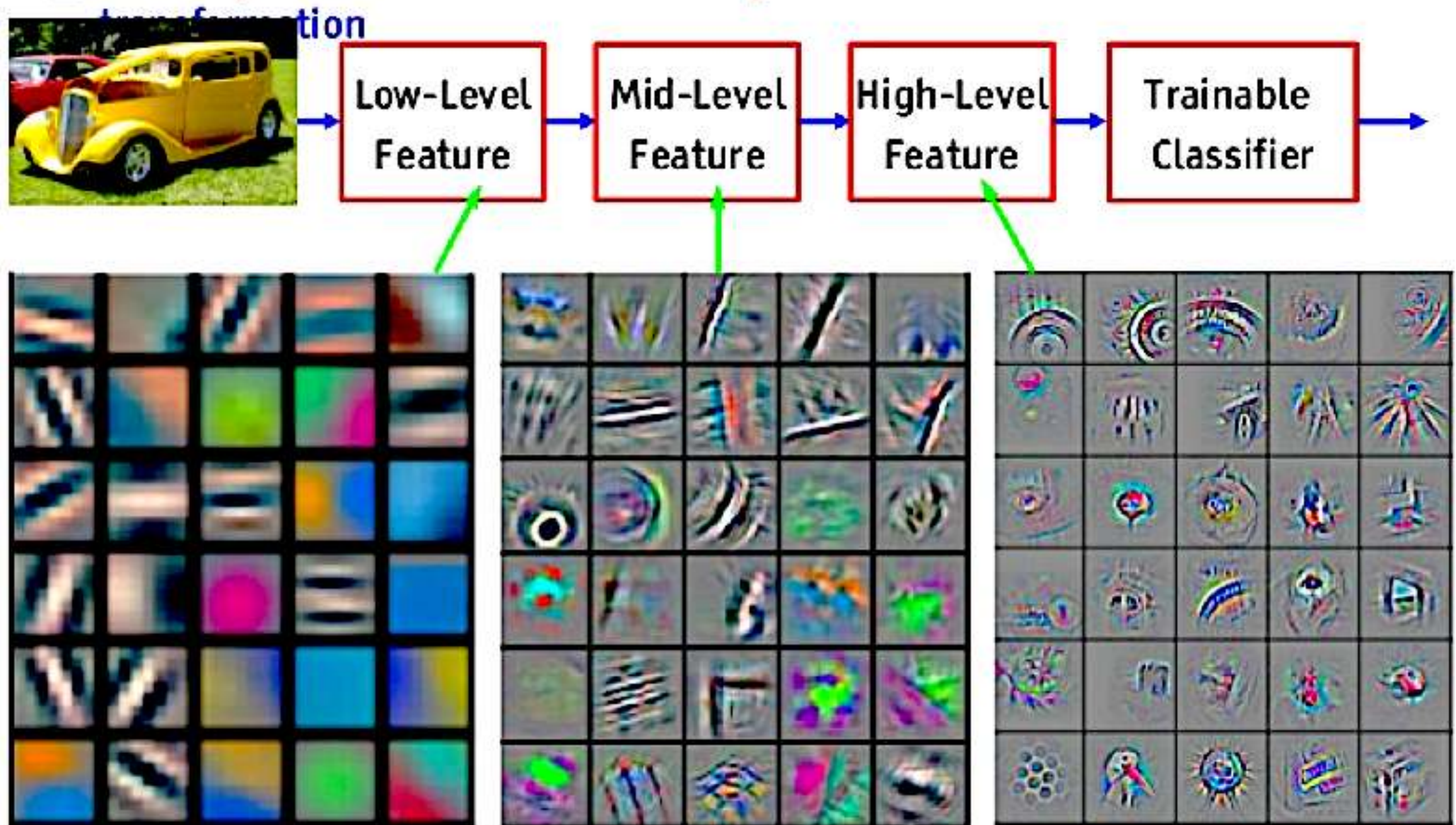
- Multiple levels of filtering and subsampling operations.
- Feature maps are generated at every layer.

CNNs for Speech and Audio



Representation Learning in CNNs

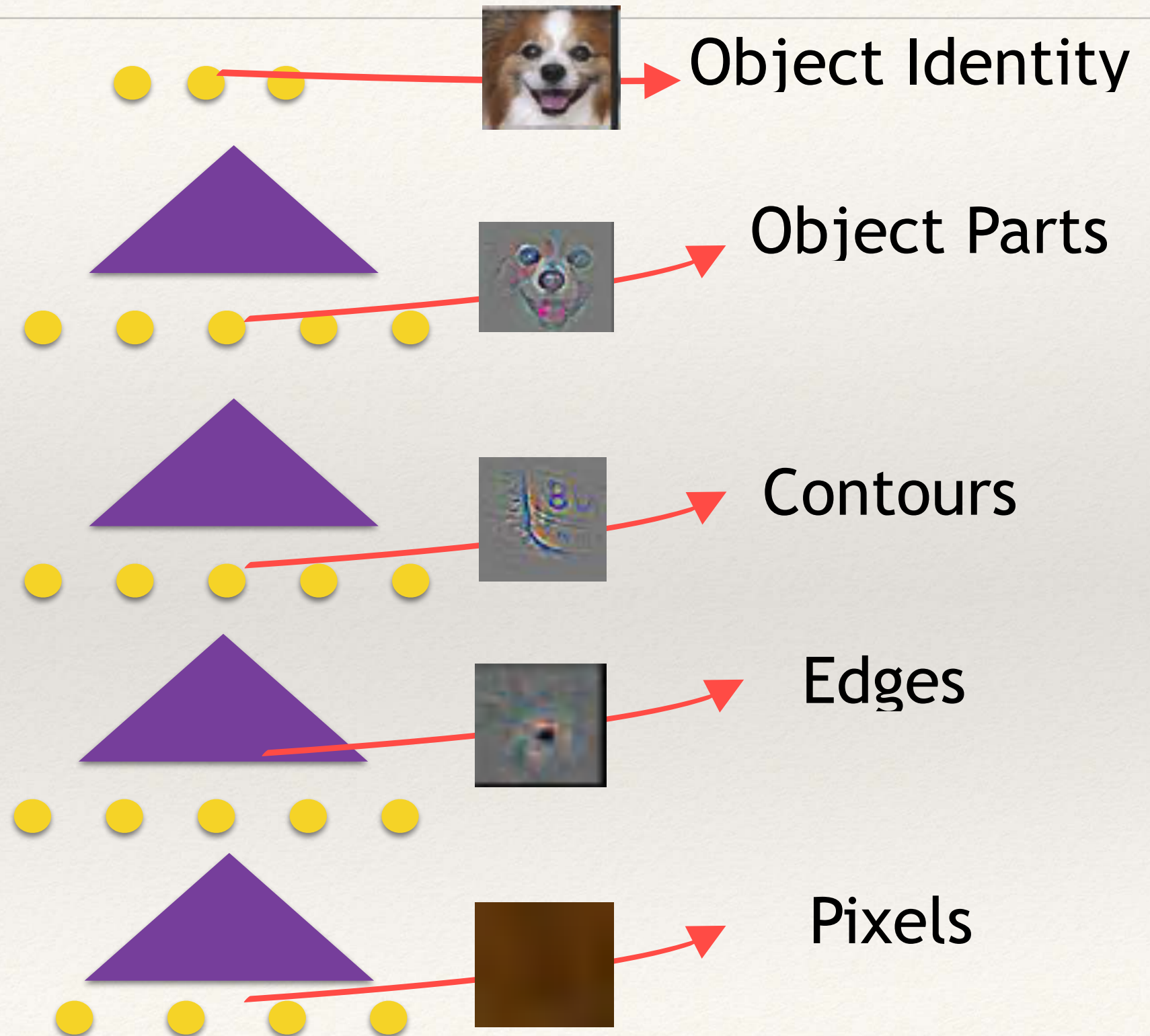
■ It's **deep** if it has **more than one stage** of **non-linear feature**



Feature visualization of convolutional net trained on ImageNet from [Zeiler & Fergus 2013]

Representation Learning in CNNs

[Zeiler, 2014]



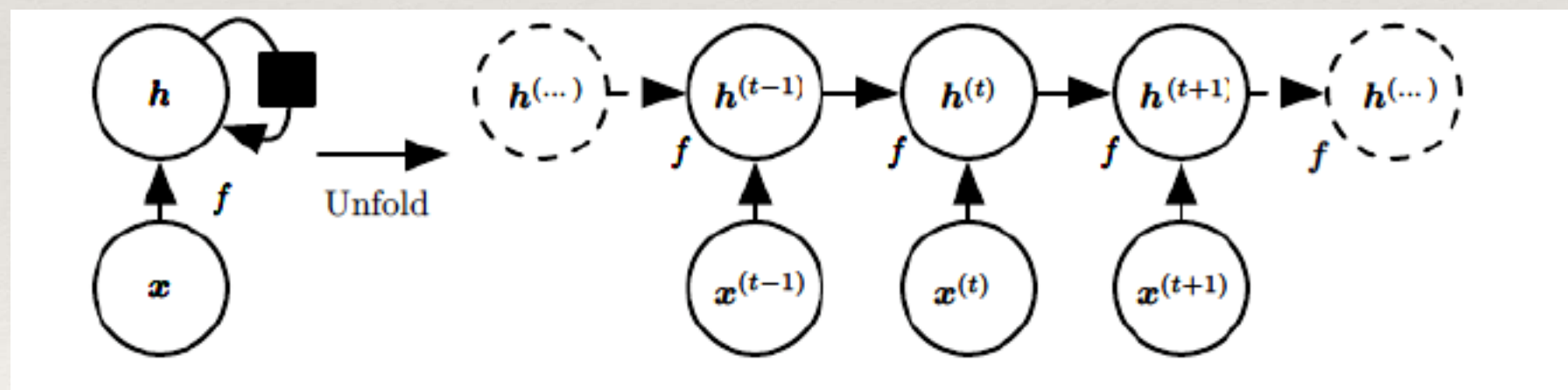
Recurrent Networks

$$s^{(t)} = f(s^{(t-1)}; \theta),$$

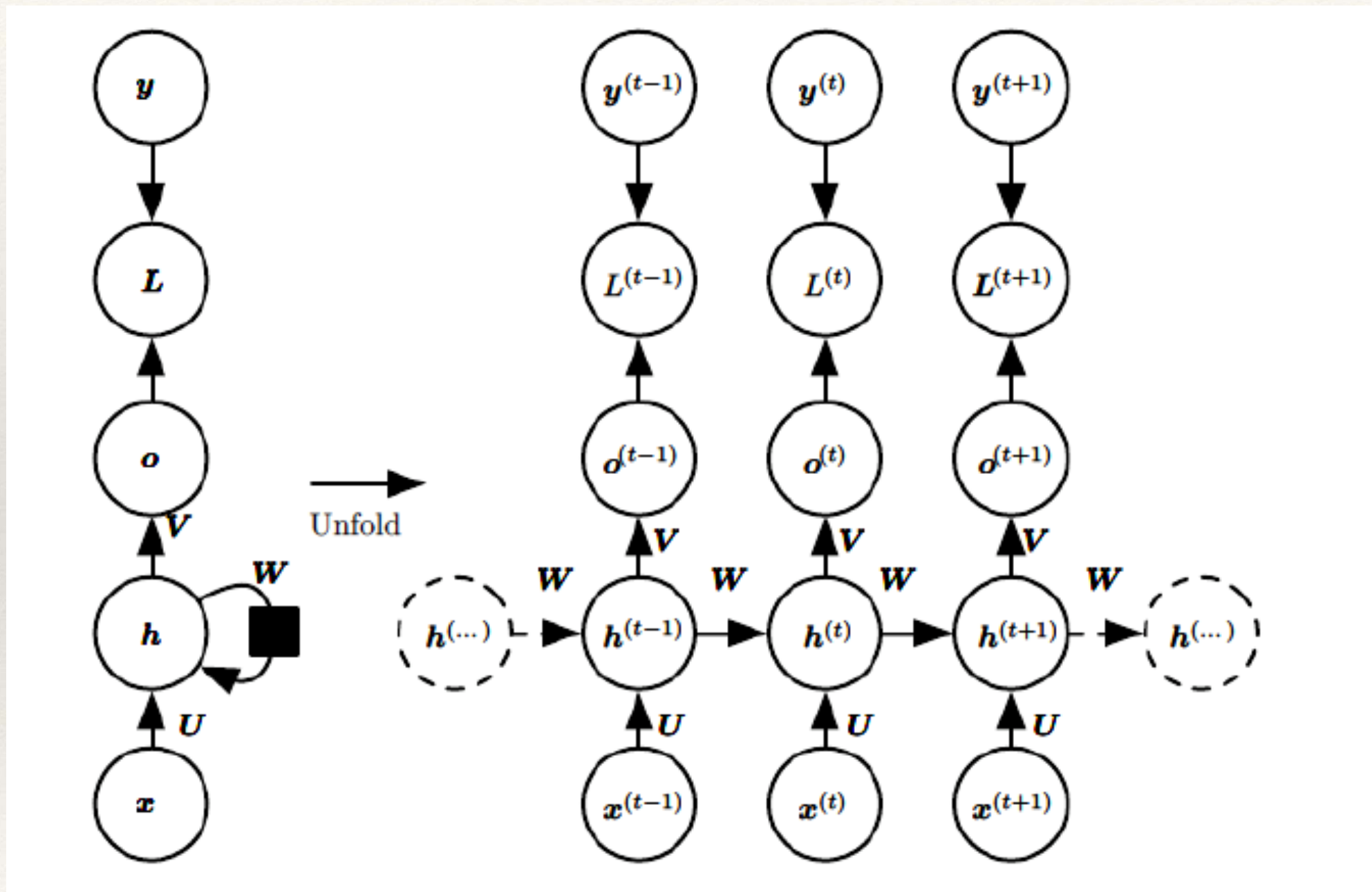
$$\begin{aligned} s^{(3)} &= f(s^{(2)}; \theta) \\ &= f(f(s^{(1)}; \theta); \theta) \end{aligned}$$

$$s^{(t)} = f(s^{(t-1)}, x^{(t)}; \theta),$$

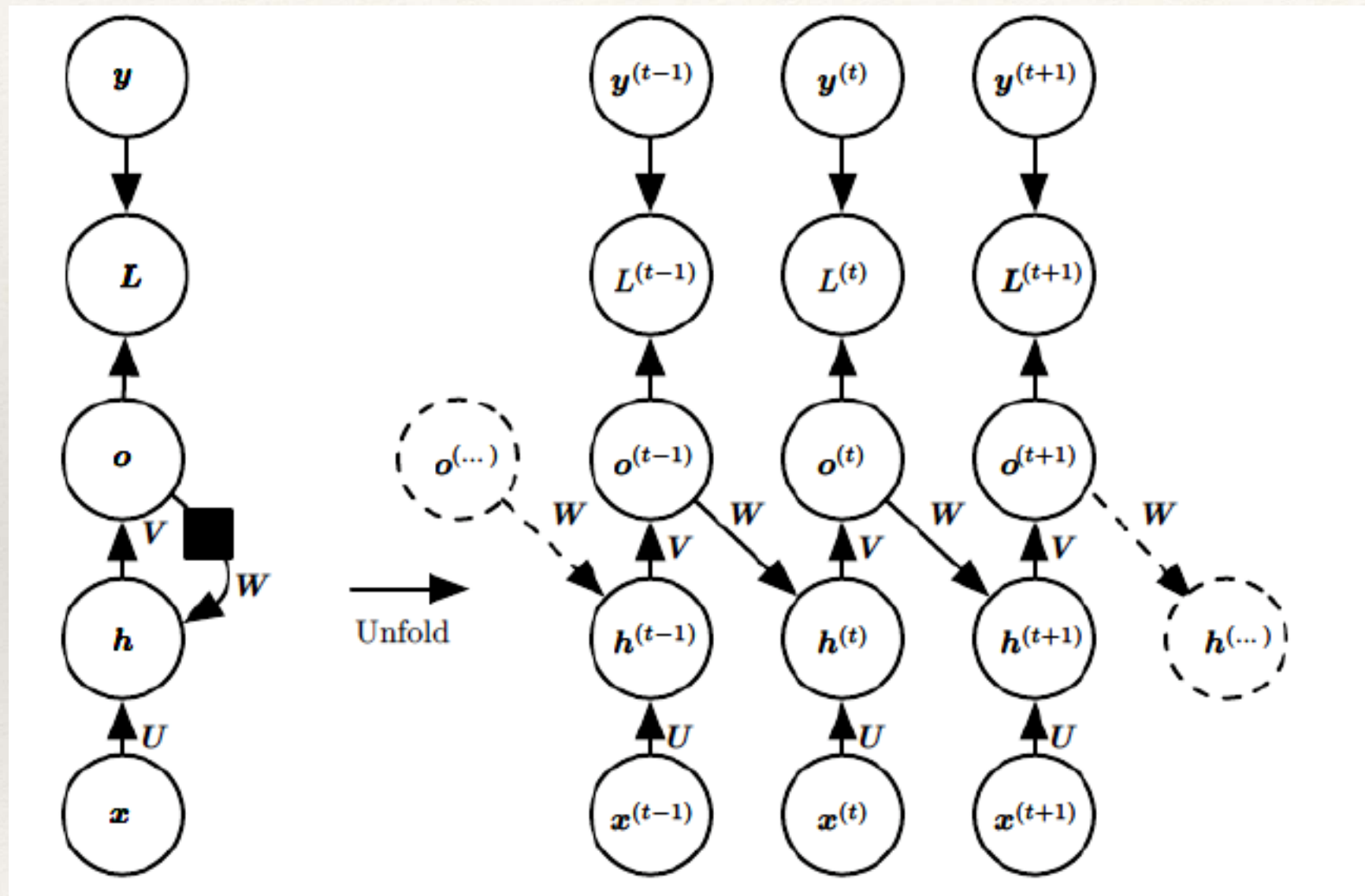
$$h^{(t)} = f(h^{(t-1)}, x^{(t)}; \theta),$$



Recurrent Networks

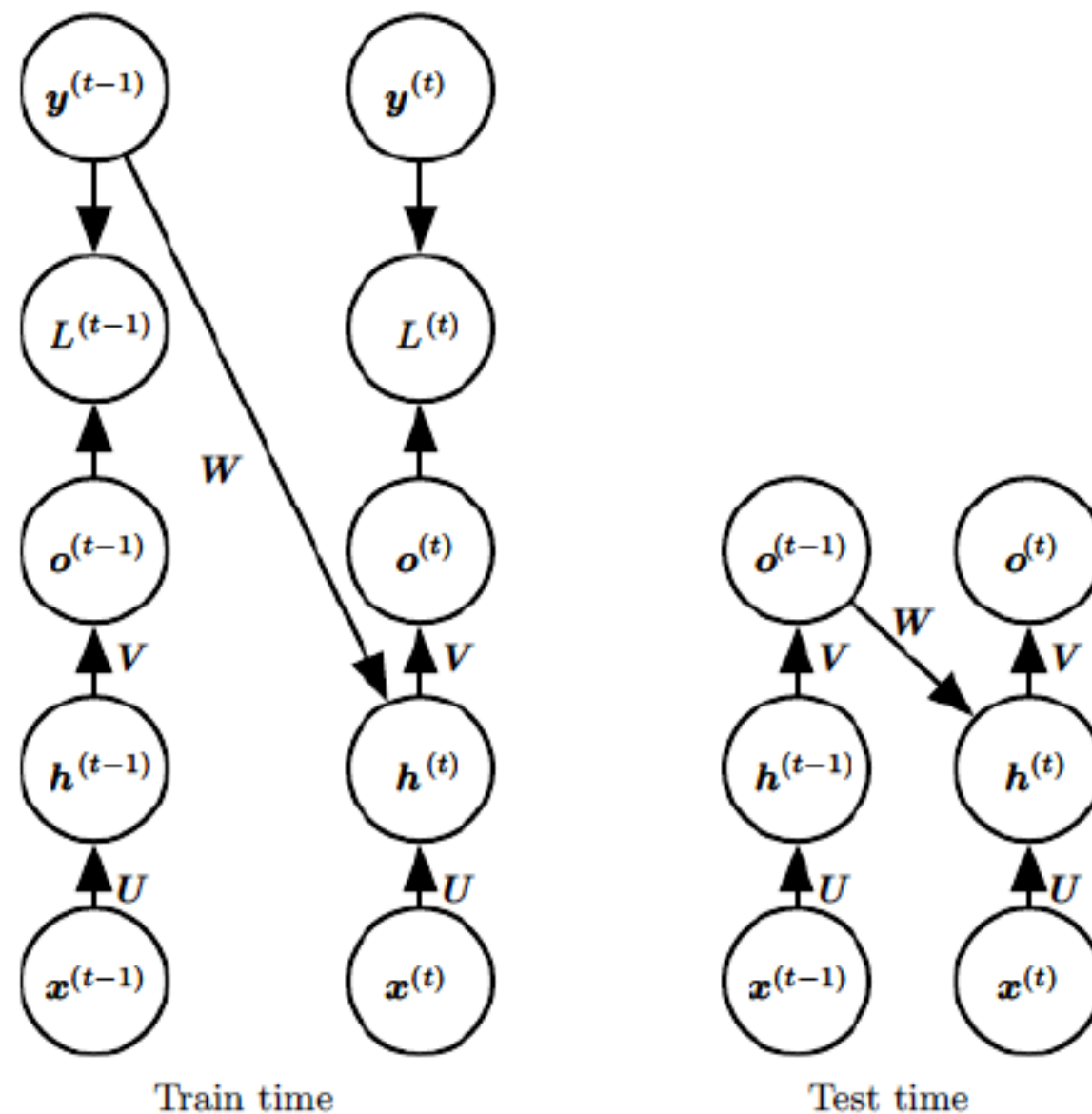


Recurrent Networks



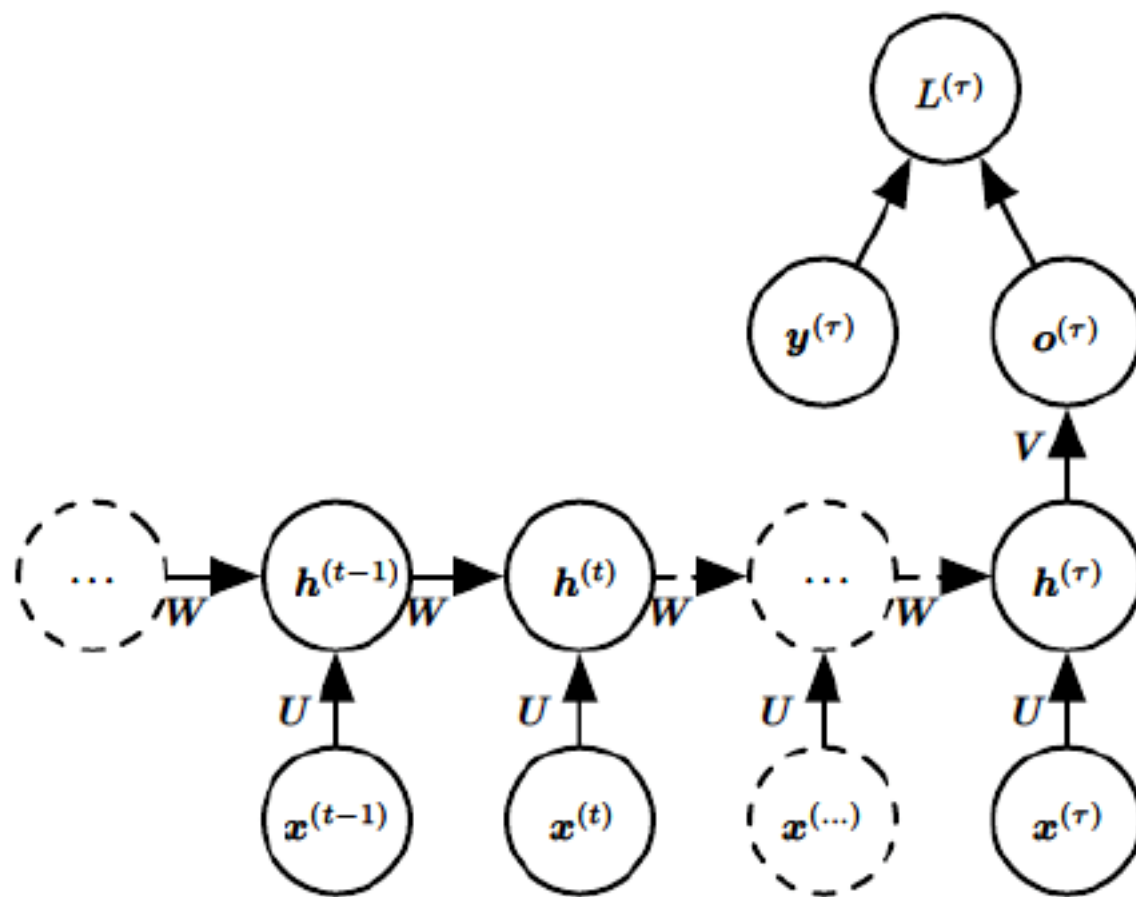
**Teacher
Forcing Networks**

Recurrent Networks



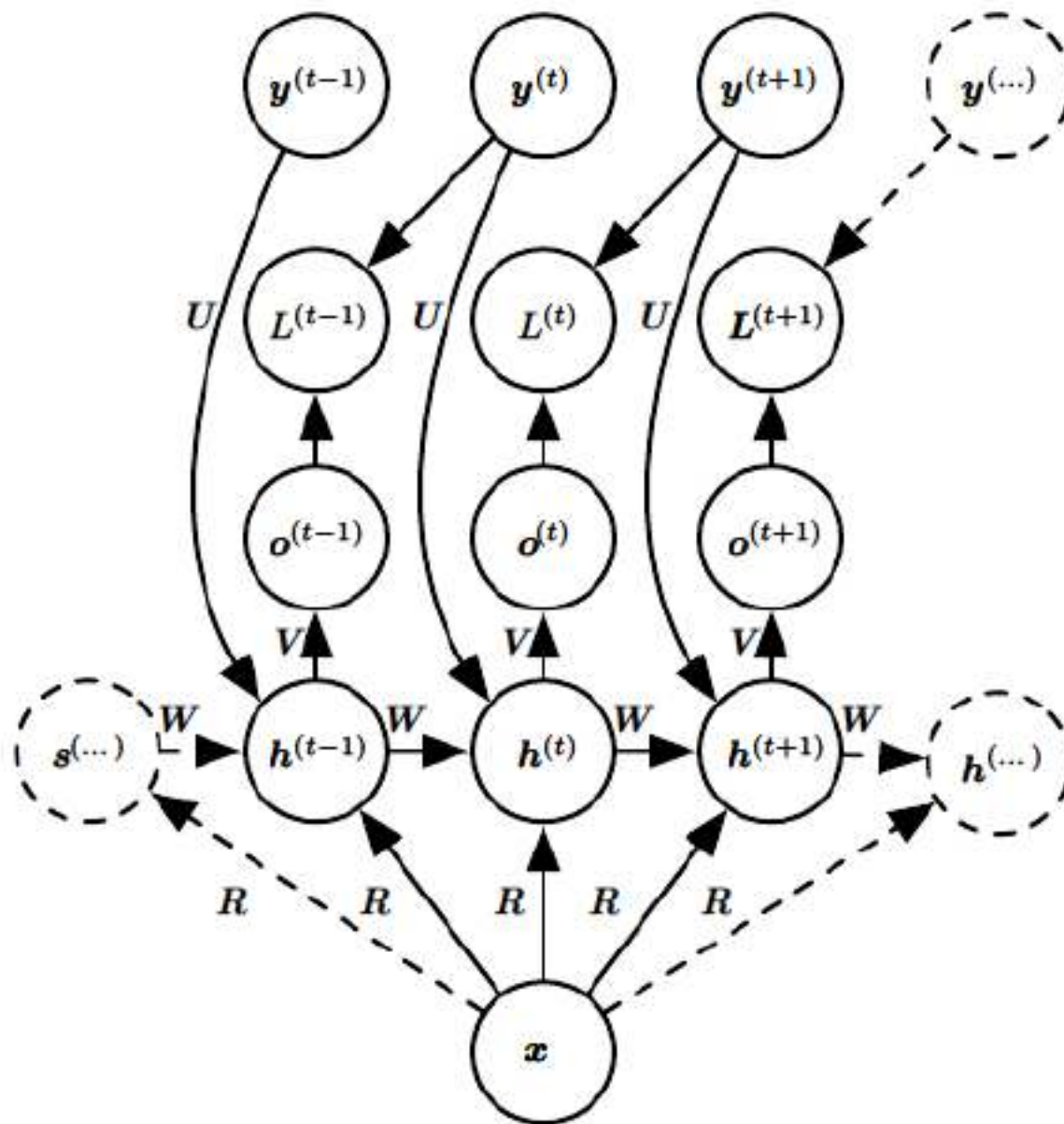
**Teacher
Forcing Networks**

Recurrent Networks



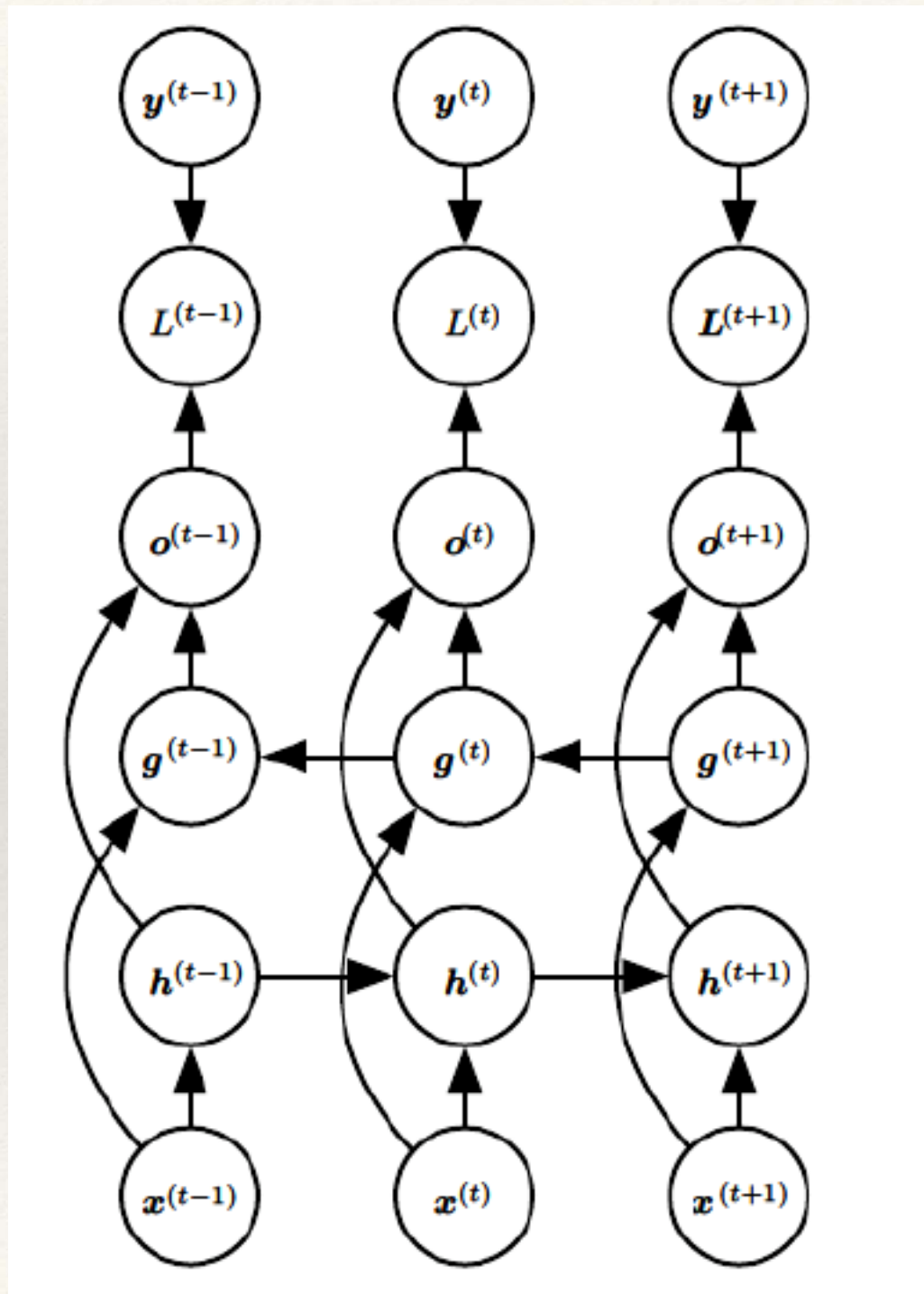
**Multiple Input
Single Output**

Recurrent Networks



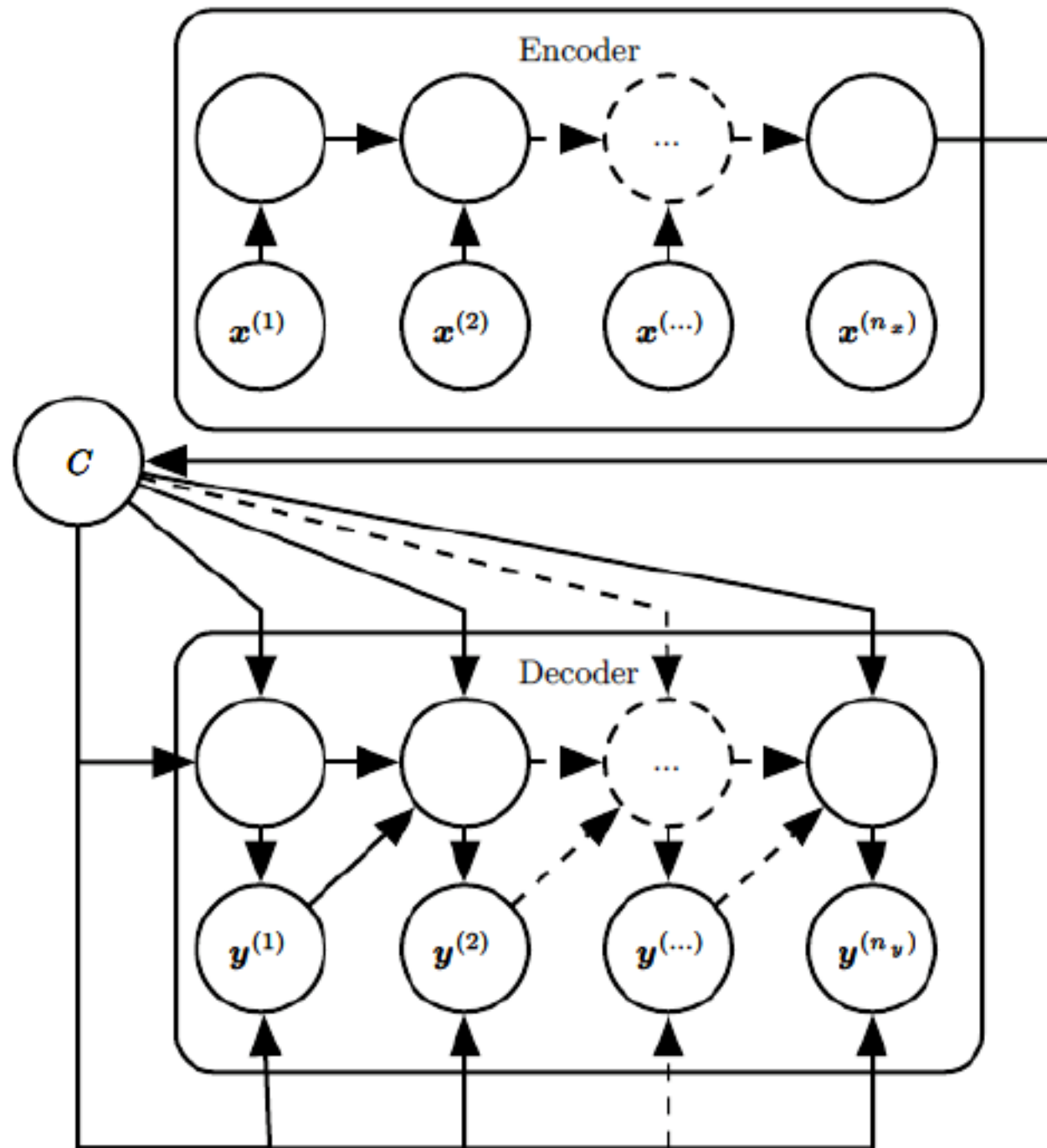
**Single Input
Multiple Output**

Recurrent Networks



**Bi-directional
Networks**

Recurrent Networks



**Sequence to
Sequence
Mapping Networks**